Lirui (Leroy) Wang

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Education:

University of Washington, Seattle, WA (GPA:**3.97**/**Annual Dean's list**)

- Bachelor of Science, Electrical Engineering (Honor), College of Engineering — Expected Jun 2020
- Bachelor of Science, Computer Science (Honor), College of Arts and Sciences — Expected Jun 2020
- Bachelor of Science, Mathematics (Minor), College of Arts and Sciences — Expected Jun 2020

Work Experience:

Intern, DJI Technology, Shenzhen, China — Jun 2017 – Aug 2017

- Worked on automatic robot battle of vehicles with TX series onboard and DJI Manifold with Guidance
- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
- Applied amcl to navigate and localize, and built a top-level state machine to strategize the robot

Research Projects:

Grasp Planning – **UW Robotics and States Estimation Lab** — Dec 2018 – Present

- Apply package GraspIt! and RMP to a table-top object pick up tasks
- Experimented with new learning-based approach that synthesize grasp pose based on shapes and geometry losses
- Prototyped RL and MPC with Pybullet to map observed points directly to 6D pose increments in task space
- Attempted residual policy by a neural net to imitate the MPC behavior
- Jointly optimize motion and grasp planning by extending the CHOMP framework
- Research extensively on traditional methods and data-driven methods in grasp synthesis

DeepIM Tracking – **UW Robotics and States Estimation Lab** — June 2018 – Dec 2018

- Tested extensively DeepIM on YCB datasets and achieves results trained on synthetic data only
- Improvednetwork training stability, pose distribution and occlusion generation
- Applied DeepIM in real-time tracking for occluded scene in YCB and AR demos
- Worked on kinematics to build generalized vision system that can track robot arm

Depth and Flow – **Computer Vision Course Project** — March 2018 – June 2018

- Investigated some deep learning methods results for depths and flow predictions
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting

SoundScape – **UW Adaptive Computing Machines and Emulator Lab** — March 2018 – June 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle

Manipulation – **UW Robotics and States Estimation Lab** — Aug 2017 – March 2018

- Solved a manipulation task with PoseCNN, which enables Baxter to manipulate detected objects
- Studied camera models, robot kinematics, Baxter SDK and ROS packages for tracking and control

Leadership and Involvement:

Advanced Robotics at the University of Washington – Computer Vision Lead & Drone Lead — Sep 2016 – Feb 2018

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI.
- Perform system integration of controls, navigation and serial communication
- Implement the detection, shooting, refilling, landing and communication using DJI SDK on DJI M100 drone

Skills:

Languages: Java, C++/C, Python, Matlab
Platform & Frameworks: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
Tools: Git, CMake, OpenCV, DJI_Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA
Selected Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing